

Total Space Complexity = Input space + Auxiliary Space.

Input space : Input data on which the task of searching/sorting needs to be done.

Auxiliary Space : It is the extra space/extra variables required to complete the task of searching/sorting.

In linear search for the arr[9] input, we need key and loop variable to complete the search.

The auxiliary space remains same, irrespective of the input space, hence we can say, the auxiliary space is constant.

Input space : input array : $O(n)$: n is the no. of elements

Aux space : extra space required to complete the task.

As the auxiliary space is constant we say, it is $O(1)$.

For space complexity, majority of the times, only auxiliary space is considered, as input space is the minimum requirement to complete any task.

So, **Space complexity : $O(1)$ for linear search**

Total Space complexity : input + aux
 $O(n) + O(1)$
 $O(n+1)$
 $O(n) \rightarrow +1$ is discarded.

total space = input space + Aux space.

X input $\rightarrow$ int arr[9] $\rightarrow$ $O(n)$

Aux $\rightarrow$ Extra variables.

✓ linear search $\rightarrow$ key, i (loop variable) $\rightarrow$

✓ binary search $\rightarrow$ key, left, right, mid.

* Aux $\rightarrow$ constant $\rightarrow$ $O(1)$.

space comp $\rightarrow$ $O(1)$
 $O(1)$

Total space comp $\rightarrow$ input space + Aux space.

$$O(n) + O(1)$$

$$\underbrace{O(n+1)}$$

$$\text{Total space comp} \rightarrow \underline{\underline{O(n)}}$$